

EFFECTS OF ELEVATED WATER TEMPERATURE ON THE CRITICAL SWIM SPEEDS OF YEARLING RAINBOW TROUT, *SALMO GAIIRDNERI**

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(Received 11 February 1982; accepted 2 June 1982)

Abstract—Effects of elevated water temperature on the critical swim speeds of rainbow trout, *Salmo gairdneri*, were investigated. Trout acclimated to 10°C were exposed to 10, 15, 20 and 25°C while swimming and at rest. Initial swim speed of 20 cm s⁻¹ was increased in 10 cm s⁻¹ increments every 20 min until the fish fatigued. Critical swim speeds were calculated in absolute values (cm s⁻¹) and relative performance values (body lengths s⁻¹). Critical swim speeds were similar at 10, 15 and 20°C. Swimming performance was significantly decreased at 25°C. Performance measured as critical swim speed was unaffected by temperature elevations up to 10°C above acclimation temperature of 10°C.

INTRODUCTION

FISH swim performance provides an integrated, total organism measurement of response to environmental change. Swim performance has been suggested as the best means to evaluate sub-lethal stress (Cairns, 1966). Swim performance is usually separated into 3 categories: sustained, prolonged and burst speeds. Sustained speeds are those that a fish can maintain for long periods. Metabolism to support this activity is primarily aerobic. Prolonged swim speeds are faster than sustained, but are maintained for less time, and are dependent on both aerobic and anaerobic metabolism. Burst speeds are the maximum absolute attainable speed and are usually maintained for seconds (Farlinger & Beamish, 1977).

Measurement of fish performance in any of the 3 categories provides an indication of swimming ability and different ecological inferences can be made from each category. Prolonged swim speed is the measurement of choice as an indicator of sub-lethal stress (Brett, 1964). Brett defined a specific measurement within the prolonged category, i.e. the critical swim speed (CSS). The CSS is the maximum velocity a fish can maintain for a specified period (less than 200 min) and is determined at the point of physical fatigue when all physiological systems fail simultaneously (Jones, 1971). Critical swim speeds have been determined with various species to measure stamina limits and stress; salinity (Glova & McInnery, 1977); hypoxia and anemia (Jones, 1971); toxicants (Peterson, 1974); and temperature (Brett, 1964, 1967; Hocutt, 1973; Otto & O'Hara-Rice, 1974).

The swim performance of several species, including rainbow trout, *Salmo gairdneri*, have been investigated. However, the effect of water temperature on the CSS of rainbow trout has never been determined.

Therefore, we investigated the swim performance of 10°C acclimated trout at 10, 15, 20 and 25°C.

MATERIALS AND METHODS

Yearling rainbow trout, *Salmo gairdneri*, were reared at the Pacific Northwest Laboratory (Richland, Washington) in flowing Columbia River water in 12 m dia. concrete ponds under ambient light and temperature out-of-doors. Two months before the study, fish were moved into the laboratory to 1.8 m dia. insulated, fiberglass tanks and the water temperature was adjusted 1°C/day to reach 10°C acclimation. Temperature was manually controlled within ±0.5°C of acclimation. Four weeks before tests, fish were moved to a trap tank (Schneider *et al.*, 1977) that permitted serial removal of individual fish without stress to the remaining stock. Water temperature in the trap tank was controlled to 10 ± 0.1°C.

Laboratory light simulated a natural 10:14 h light-dark cycle. Artificial sunrise and sunset were provided over 30 min. Lighting was a combination of "cool white" fluorescent and incandescent bulbs. Fish were fed a dry pellet commercial fish food. Feeding level was 1.5–1.75% of mean body weight of stock per day. Average weight and mean standard length was 495.6 ± 81.0 g and 28.7 ± 1.5 cm, respectively.

We met the specific requirements of this technique with a specially designed and constructed swim tunnel similar to that described by Brett (1964). The tunnel (exerciser) consisted of a 25 cm dia. tube constructed of polyvinyl chloride and polystyrene pipe fashioned in a closed loop. The exercise area of the loop was 2.2 m long. A chamber on a side arm of the loop permitted holding a fish quiescent (the control) while a second fish was undergoing forced exercise. Water velocity and temperature were controlled automatically by a Research Inc. Model 70168 Data Trak electrostatic curve following programmer. This allowed replicate experimental programs. Water was propelled

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by a two-stage jet pump and water temperature was maintained by mixing heated and chilled water.

Swim performance was measured in Columbia River water at 4 test temperatures: 10, 15, 20 and 25°C. The exerciser was filled until the swim chamber was about half full. One fish from the trap tank was placed in the chamber, the loading port closed and the filling process continued. During this procedure, the jet pump produced a current velocity of 20 cm s⁻¹. When the loading port closed, a second fish from the trap tank was placed in the holding chamber as the resting or control fish. The exerciser was then filled and purged of all air. Initial water temperature in the exerciser was 10°C for test at acclimation temperature or 13°C for tests at 15, 20 or 25°C. Initial current velocity (20 cm s⁻¹) remained constant for 30 min after placing fish in the exerciser. This time period was provided to adjust water temperature to test conditions, allow fish to recover from handling and become accustomed to the exerciser, and, in the case of the exercised fish, to orientate to the current. At 30 min elapsed time, water velocity in the exerciser was increased in 10 cm s⁻¹ increments every 20 min until the swimming fish fatigued.

Electric shocks discouraged test fish from resting on the tail screen at the downstream end of the exerciser chamber. When a fish no longer responded to the shocks or as more usual, fatigued and was swept against the screen, the time was noted and the test terminated. Both exercised and control fish were recovered from the exerciser, weighed and measured.

Critical swim speeds were calculated with the formula described by Brett (1964) and Farlinger & Beamish (1977) and recorded in absolute values of cm s⁻¹ and converted to relative performance, body lengths per second (BL s⁻¹). Weight and length measurements were analyzed by a two-way analysis of variance. The 2 factors in the analysis were treatment (resting or exercise) and temperature (10, 15, 20 and 25°C). The CSS were analyzed in a one-way analysis of variance against test temperature.

RESULTS

Control and exercising fish were routinely observed throughout each test and a typical behaviour pattern was noted. Fish were initially excited and struggled in the apparatus. Fish in the control chamber quickly became quiescent, usually resting dorso-ventrally upright on the chamber floor. Fish in the exercise chamber often made several runs upstream and downstream in a head down position as though attempting to sound. The fish usually quietened to a steady swimming, mid-bore in the chamber before the first speed change, i.e. to 30 cm s⁻¹. The anterior two-thirds of the swim chamber was draped with black plastic to avoid additional stimulation of the swimming fish. About 0.5 m of the downstream portion of the chamber was left uncovered to observe fish near the tail screen. Exercising fish typically remained in the covered end of the chamber until shortly before fatigue. Simultaneous with fatigue fish turned colour from silvery to a dark green-black. Additionally, fatigued fish attempted to gain advantage against the current by resting their tails on the electrified screen and their side on the chamber wall. Tail screen shocks

were not needed until current velocity approached 50 cm s⁻¹ and were usually required above 80 cm s⁻¹.

Both control and exercise fish displayed signs of thermal stress during the 25°C tests. A cough response and release of air bubbles from the mouth and opercles were frequently noted in exercising fish. Furthermore, erratic swimming, bursts and shuddering movements were observed. Similarly, resting fish became agitated soon after test temperature was reached. Interestingly, exercising fish appeared to tolerate 25°C better than controls. Ten tests were conducted at 25°C and, of these, 8 control fish lost equilibrium. One control fish died before the test termination. Although some exercising fish lost equilibrium briefly, they recovered and continued swimming effort.

A statistically significant effect of test temperature was determined in the swim performance of rainbow trout. Mean CSS at 25°C was 47.95 ± 12.39 cm s⁻¹, which is significantly lower ($P < 0.05$) than the recorded CSS at 10°C (89.93 ± 12.33 cm s⁻¹), 15°C (94.82 ± 11.11 cm s⁻¹) or 20°C (93.23 ± 8.73 cm s⁻¹). Similar results were obtained for relative swim speed. Mean swim speed at 25°C was 1.64 ± 0.42 BL s⁻¹; significantly less than the 10°C (3.15 ± 0.40 BL s⁻¹), 15°C (3.37 ± 0.39 BL s⁻¹), and 20°C (3.21 ± 0.33 BL s⁻¹) swim speeds. There was no difference in performance at the 3 lowest test temperatures. The results are illustrated in Fig. 1. There were no significant differences ($P < 0.05$) between weight or length means for control and exercised fish at any test temperature. There was no significant difference among weight or length means among test temperatures either.

DISCUSSION

We recorded CSS's in both absolute (cm s⁻¹) and relative (BL s⁻¹) units, Fig. 1. Because various sizes and species of fish were used by previous investigators, the latter unit is more appropriate for comparative purposes. Swim speeds obtained by us compare

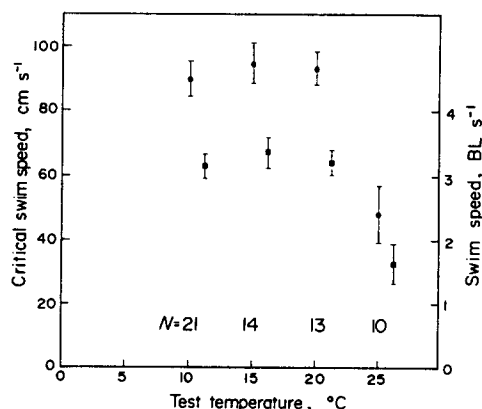


Fig. 1. Plot of rainbow trout, *Salmo gairdneri*, critical swim speeds in absolute (cm s⁻¹, ○) and relative values (body lengths (BL) s⁻¹, ●), and the 95% confidence intervals measured at 10, 15, 20 and 25°C. Sample size is shown as N.

favourably with those observed by others investigating various aspects of salmonid swim performance. At 10, 15 and 20°C, mean relative swim speed ranged from 3.15 to 3.37 BL s⁻¹. Blaxter (1964) reported that salmonid swim speeds typically range from 3.0 to 4.0 BL s⁻¹. Data from Brett's (1964) early studies of sockeye salmon swim speeds agree with this range: he reported 4.0 BL s⁻¹ at acclimation temperature.

At a test temperature of 25°C (15°C above acclimation) we observed a significant decrease in CSS value. Similar results have been reported for other species where for any given acclimation temperature a plot of CSS versus temperature takes on a "dome" shape (Brett, 1967). Peak swim performance is usually at a temperature close to the species preferendum (Fry & Hart, 1948). The dome shape curve is usually flattened within 5 to 10°C of this temperature (Brett, 1964). Our results showing no significant difference in CSS at 10, 15 and 20°C support this view. The ultimate preferendum for rainbow trout lies within this temperature range (Javaid & Anderson, 1966).

Critical swim speed has been supported by numerous authors as an indicator of response to stress. Sprague (1971) indicated that swimming ability was the single best measure of performance. Our data indicate, therefore, that swim performance of rainbow trout acclimated to 15°C is relatively insensitive to temperature elevations up to 10°C. No difference was observed in CSS measured in this range. However, when test temperature places fish in the zone of thermal resistance, i.e. 25°C, their performance is halved. Although this temperature was approximately 2°C below the upper incipient lethal temperature for rainbow trout (Charlon *et al.*, 1970), an acute effect is apparent in the ability of the species to maintain swim performance.

Acknowledgements—This work was sponsored by the U.S. Department of Energy under contract number DE-AC06-76RLO-1830. The authors wish to express their appreciation to Messrs R. G. Genoway and S. A. Barracough for their technical assistance in conducting the study and to Drs C. D. Becker and R. H. Gray for reviewing the manuscript.

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Key Word Index—Water temperature; critical swim speed; exercise; fatigue; stamina; trout; *Salmo gairdneri*; acclimation.